



Calculus & Analytical Geometry

Section A: Power Rule (Polynomial Functions)

1. AI Model Training Load

An AI model's computational load (in tera-operations) during training is modeled by

$$C(t) = 5t^4 - 3t^2 + 7t$$

where t is time in hours. Find the rate of change of computational load at time t .

2. Game Rendering Complexity (MMG)

The rendering complexity of a 3D game scene is given by

$$R(x) = 2x^5 - 7x^3 + x$$

where x is the number of active objects (in hundreds). Find $R'(x)$.

Section B: Constant Multiple Rule

3. Cyber Attack Detection Score

A threat detection score is defined as

$$S(t) = 12(3t^3 - 5t + 4)$$

where t is time in seconds. Find the rate at which the score changes with respect to time.

4. AI Data Processing Speed

The data processed by an AI system is modeled as

$$D(x) = -8x^4 + 6x^2$$

where x is the number of processing cycles (in thousands). Differentiate $D(x)$.



Worksheet#02: Differentiation Rules Part-01

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Section C: Sum and Difference Rule

5. Network Traffic Load (Cyber Security)

Total network traffic load is given by

$$T(t) = (4t^3 + 6t^2) - (3t^2 - 5t)$$

where t is time in minutes. Find $T'(t)$.

6. AI Memory Consumption

The memory consumption of an AI algorithm is modeled by

$$M(x) = (7x^4 - 2x^2) + (5x^3 - 9x)$$

Find the rate of change of memory consumption.

Section D: Product Rule

7. Game Physics Engine (MMG)

The force applied to a character in a game is given by

$$F(t) = (3t^2 + 2t)(t^3 - 4)$$

where t is time in seconds. Find $F'(t)$.

8. AI Signal Processing

An AI signal strength function is modeled as

$$A(x) = (5x^2 - 1)(2x^3 + x)$$

Find the derivative of $A(x)$.

Section E: Quotient Rule

9. Encryption Efficiency (Cyber Security)

The efficiency of an encryption algorithm is modeled by

$$E(x) = \frac{4x^3 - 2x}{x^2 + 1}$$

where x is the key size parameter. Find $E'(x)$.

10. Frame Rate Optimization (MMG)

The frame rate of a game is given by

$$F(x) = \frac{6x^2 + 3x}{x^2 - 1}$$

where x represents GPU clock cycles. Differentiate $F(x)$.



Worksheet#02: Differentiation Rules Part-01

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Section F: Velocity, Acceleration & Horizontal Tangents

11. AI Robot Motion (Velocity & Acceleration)

The position of an AI-powered robot is given by

$$S(t) = t^4 - 6t^2 + 4t$$

(a) Find the velocity function.

(b) Find the acceleration function.

12. Network Packet Flow (Cyber Security)

The displacement of a data packet is modeled as

$$C(t) = 2t^3 - 9t^2 + 12t$$

Find:

(a) Velocity at time t

(b) Acceleration at time t

13. Game Object Movement (MMG)

The position of a moving object in a game world is

$$S(t) = -t^3 + 6t^2 - 9t$$

Find the velocity and acceleration functions.

14. Horizontal Tangent – AI Cost Function

The cost function of an AI model is

$$C(x) = x^4 - 8t^2 + 3$$

Find the points on the curve where the tangent line is horizontal.

15. Horizontal Tangent – Cyber Security Risk Curve

A risk assessment curve is given by

$$R(x) = 2x^3 - 9x^2 + 12x$$

Find the points where the tangent is horizontal.

16. Horizontal Tangent – Game Optimization Curve

The performance curve of a game engine is

$$P(x) = x^5 - 5x^3 + 4x$$

Find all points on the curve where the tangent line is horizontal.



Worksheet#02: Differentiation Rules Part-01

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Section G: Mixed Scenario-Based Derivative Models (Advanced)

17. AI Model Computational Cost (Mixed Polynomial + Constant Multiple)

The computational cost of training an AI model is given by

$$C(x) = 7(2x^4 - 3x^2 + 5x) - (4x^3 - x)$$

Find the rate of change of computational cost with respect to x .

18. Cyber Security Network Load (Sum & Difference + Polynomial)

The total network load in a cyber system is modeled as

$$N(t) = (5t^3 - 2t^2 + 6t) + (3t^4 - 7t^2) - (2t^3 - t)$$

Find $N'(t)$.

19. Game Rendering Performance (Product Rule)

The rendering performance of a game engine is modeled by

$$P(x) = (x^2 + 3x)(4x^3 - 5)$$

Find the derivative of the performance function.

20. AI Signal Optimization Ratio (Quotient Rule)

The AI signal-to-noise optimization function is

$$S(x) = \frac{3x^4 - 2x^2 + 1}{x^2 + 2x + 1}$$

Find $S'(x)$.

21. Cyber Attack Response Efficiency (Mixed Product + Sum)

The efficiency of a cyber response system is given by

$$E(t) = (2t^3 + t)(t^2 - 4t + 1)$$

Find the derivative of the efficiency function.



Worksheet#02: Differentiation Rules Part-01

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Section H: Advanced Mixed Models (Polynomial + Radical + Exponential)

22. AI Hybrid Learning Model (Polynomial + Exponential)

The learning performance of an AI system is modeled by

$$L(x) = (3x^2 - 4x)e^{0.5x} + 2x^3$$

Find $L'(x)$.

23. Cyber Security Risk Signal (Quotient + Radical)

The cyber risk signal index is given by

$$R(x) = \frac{4x^{\frac{3}{2}} - 2\sqrt{x}}{x^2 + 1}$$

Find the derivative using the quotient rule.

24. Game Physics Energy Model (Product + Radical)

The energy of a game object is modeled as

$$E(x) = (2x^2 + 3x)\sqrt{x}$$

Find $E'(x)$ using the product rule.

25. AI Signal Amplification (Polynomial + Exponential)

The signal amplification in an AI chip is given by

$$A(t) = (t^3 - 2t)e^{0.3t}$$

Find the derivative of the amplification function.

26. Cyber Throughput Efficiency (Mixed Sum + Exponential + Polynomial)

The throughput efficiency of a secure network is modeled by

$$T(x) = 5x^2 + 3e^{0.4x} - 2x^{\frac{1}{2}}$$

Find the rate of change of throughput efficiency with respect to x .

Tangent and Normal line Problems

27. Find equations of the tangent line and normal line to the curve at the given point $y = x^4 + 2e^x$, $(0,2)$. Illustrate by graphing the curve and these lines.

28. Find equations of the tangent line and normal line to the curve at the given point $y = x^3 + e^x$, $(0,1)$. Illustrate by graphing the curve and these lines.

29. Find the points on the curve $y = x^3 - 3x$, where tangent line is horizontal.

30. Find the points on the curve $y = x^4 - 4x^2$, where tangent line is horizontal.

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